

Jeremy Lopez

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EDUCATION

University of Central Florida

Bachelor of Science in Mechanical Engineering — GPA: 3.3

Expected Graduation: 2028

EXPERIENCE

Research Assistant — Interfacial Heat Transfer Laboratory, UCF

June 2026 – Present

- Design and validate microchannel geometry for a pulsating heat pipe (PHP) intended for high-heat-flux ($\geq 100 \text{ W/cm}^2$) power electronics cooling, under the guidance of faculty advisor Dr. Shawn Putnam.
- Research and synthesize peer-reviewed literature and experimental data on flat-plate pulsating heat pipes — covering working fluid behavior, condenser configuration, and channel geometry — to ground design decisions in validated results.
- Develop SolidWorks CAD models of the microchannel heat exchanger and prepare technical presentations and documentation communicating design rationale, governing equations, and thermal performance data to faculty.

Resident Assistant — UCF Housing & Residence Life

May 2026 – Present

- Support a community of students by fostering a safe, inclusive, and academically focused living environment.
- Serve as a first point of contact, providing guidance, conflict resolution, and referrals to campus resources.
- Partner with housing staff to plan and execute community programs and enforce residential policies.

Drive Team Lead — ASME Student Design Competition

Sept 2025 – April 2026

- Developed a SolidWorks CAD model for a competition-compliant waste-sorting robot integrating color sensors.
- Integrated electronics — including motors, controllers, and power systems — with mechanical design, validating performance through cross-disciplinary team testing.
- Gained hands-on experience in cross-disciplinary teamwork, project management, and competition.

PROJECTS

Intro to Engineering – Great Navel Orange Race

Jan 2026 – April 2026

- Designed and fabricated an autonomous pond racing boat using laser-cut water-resistant wood, incorporating custom CAD-designed DXF ribs and hull structures for durability and performance.
- Integrated an Arduino Nano-based control system with a brushless motor and ESC, testing propulsion and maneuverability under competition conditions.
- 3D-printed a custom rudder assembly and waterproofed the vessel, validating environmental durability using CAD/CAM manufacturing and rapid prototyping techniques.

Arduino-Based Auto-Irrigation System

Jan 2026 – Feb 2026

- Built a low-power automated plant watering system using an Arduino Nano, soil moisture sensor, and relay-controlled pump.
- Engineered and tested a single-supply 5V power architecture with surge suppression and power stabilization, validating reliable operation under motor load.
- Modeled and 3D-printed a custom enclosure integrating a reservoir, electronics housing, and plant pot into a portable, field-tested unit.

Intro to Engineering – Catapult Project

Aug 2025 – Dec 2025

- Collaborated with a 4-member team to design and test a latch-triggered elastic catapult robot for a school-sanctioned competition.
- Implemented servo and Arduino electromechanical integration, iteratively testing to optimize launch accuracy and consistency — earned 1st place.

ACTIVITIES & LEADERSHIP

SolidWorks Student Leadership Program

Sept 2025 – Present

- Complete SolidWorks workshops and assignments while utilizing SSLP resources and practice tests to pursue industry-recognized CSWA and CSWP certifications; engage in team-based design projects and technical workflows.

UCF Brazilian Jiu-Jitsu Club

June 2025 – Present

- Active member of the university Jiu-Jitsu club, training and competing consistently while balancing academic and extracurricular commitments.
- Collaborate with teammates during training sessions to develop strategies and refine techniques, emphasizing teamwork and problem-solving.

SKILLS

Software: Excel, PowerPoint, Word, SolidWorks

Interpersonal: Communication, Leadership, Problem-Solving, Project Management, Adaptability

AWARDS & HONORS

- 2024 and 2025 AP Scholar with Distinction
- Founding Secretary, Marjory Stoneman Douglas High School Physics Club